

📌 Quick Recall (Ch. 6)

Topological planar graph & faces (§6.1). A planar graph $G = (V, E)$ together with a fixed planar drawing is called a *topological planar graph*. The connected regions of the plane lying on none of the arcs of the drawing are the *faces* of G ; the unbounded region is the *outer face*.

Map coloring (§6.4). We regard a map as a topological planar graph viewed through its faces. Two faces are *neighbors* if they share an edge on their boundaries. A k -coloring of the map assigns each face one of k colors so that neighboring faces get different colors.

Dual graph G^* (Def. 6.4.3). Vertices of G^* are the faces of G ; an edge $e \in E$ that is the common boundary of faces F_i, F_j contributes the edge $\{F_i, F_j\}$ in G^* . The map of G is k -colorable iff $\chi(G^*) \leq k$.

Handshake (Prop. 4.3.1). $\sum_{v \in V} \deg_G(v) = 2|E|$.

Face-edge double count (cf. proof of 6.3.2). Each edge appears on the boundary of exactly 2 face-sides, so

$$\sum_{F \text{ face}} |\partial F| = 2|E|,$$

where $|\partial F|$ (the *circumference* of F) is the number of edges on the boundary of F .

Euler's formula (Prop. 6.3.1). For a connected planar graph G with f faces:

$$|V| - |E| + f = 2.$$

📌 Practice: Textbook Exercise 6.4.10* — face 2-coloring of even-degree planar graphs

- (a) Consider a planar graph G with all degrees even. Prove that the map arising from (any) planar drawing of G can be colored by 2 colors.
- (b) Using (a), prove that there exists no topological planar graph with all degrees even such that all the inner faces are triangles (i.e. bounded by cycles of length 3) and the outer face is a pentagon.

💡 Hints (textbook)

Hint for (a). By induction on the number of edges. Consider a face F and the set of edges E_F lying on its boundary. Show that the degrees of all vertices in the graph (V, E_F) are even. Delete the edges from E_F , color by induction, put E_F back, and recolor the face F by the other color.

Hint for (b). Color the faces by 2 colors. The number of edges can be counted as the sum of the circumferences of the faces of one color, and also as the sum of circumferences of the faces of the other color. But one way of counting gives a number divisible by 3, the other a number not divisible by 3.

Answer: Part (a). Induction on $|E|$. (Used: all even degrees $\Rightarrow$ no bridge $\Rightarrow$ every edge separates 2 distinct faces.)

Base ($|E| = 0$). Whole plane is the only face; color red. ✓

Step. Pick any face F ; let E_F = edges on ∂F .

- (V, E_F) even degrees. No bridge $\Rightarrow \partial F$ traverses each E_F -edge once. Closed walk pairs in/out at every visit $\Rightarrow$ even degree at every vertex.
- $G' := (V, E \setminus E_F)$. $\deg_{G'} = \text{even} - \text{even} = \text{even}$; $|E(G')| < |E| \Rightarrow$ IH gives 2-coloring c' of G' -faces.
- *Lift.* In G' : $\tilde{F}$ (face containing F) = $F \cup \{F\text{'s } E_F\text{-neighbors}\}$; other G -faces unchanged.
No collision: two E_F -neighbors F_a, F_b joined by $e' \in E \setminus E_F \Rightarrow e'$ has $\tilde{F}$ on both sides in $G' \Rightarrow$ bridge in G' — but G' also has all even degrees. ✗
 Say $c'(\tilde{F}) = \text{red}$. Set $c(F) := \text{blue}$; every other face keeps its G' -color (so F 's E_F -neighbors are all red).
- *Proper.* $e \in E_F$: F (blue) vs E_F -neighbor (red). ✓ $e \in E \setminus E_F$: 2 distinct G' -faces (by no-collision) $\Rightarrow$ different c' -colors $\Rightarrow$ different c -colors. ✓

Part (b). Suppose such G exists. By (a), 2-color faces red/blue. Each edge: red on one side, blue on the other $\Rightarrow$

$$\sum_{F \text{ red}} |\partial F| = |E| = \sum_{F \text{ blue}} |\partial F|.$$

WLOG pentagon is red; every other face is a triangle.

- *Blue faces* (all triangles): $|E| = \sum_{F \text{ blue}} |\partial F| = 3 \cdot (\# \text{blue}) \Rightarrow 3 \mid |E|$.

- *Red faces* (1 pentagon + triangles): $|E| = \sum_{F \text{ red}} |\partial F| = 5 + 3 \cdot (\# \text{red triangles}) \Rightarrow |E| \equiv 2 \pmod{3}$.
 $\nmid$.

□